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FreeViM: A Patch-free Global-local Visual Mamba Framework for Multi-class Change Detection in Hyperspectral Images

Lifeng Wang, Bosheng Wei, Maofan Zhao, Yuanxin Ye, and Lorenzo Bruzzone

***Abstract*—Current methods for multi-class change detection (MCD) in hyperspectral images (HSIs) face critical bottlenecks in both accuracy and efficiency. Traditional patch-based methods suffer from redundant computation in highly overlapping patches, while their fixed patch size constrains the receptive field and limits the modeling of long-range global dependencies across the entire HSI. Moreover, existing Visual Mamba (VMamba)-based methods often lose crucial short-range dependencies as spatial tokens are flattened and processed along fixed scanning patterns, which restricts dynamic long-range global context and short-range local detail interactions. To overcome these limitations, we propose FreeViM, a patch-free global-local fast inference framework for MCD in HSIs. FreeViM integrates three key components: 1) a patch-free VMamba network that models global context without redundant overlapping-patch computations; 2) a local context clustering module that dynamically captures fine-grained local spatial details; and 3) an adaptive global-local feature interaction strategy that effectively fuses long-range global contexts with short-range local details. By bridging the gap between global receptive fields and local fine-grained dependencies, FreeViM provides a robust solution for HSI-MCD task, achieving superior performance in both accuracy and computational efficiency on Yancheng, Umatilla and Hermiston benchmark datasets.**

***Index Terms*—Multi-class change detection, global-local modeling, patch-free, vision Mamba, context clustering, remote Sensing**

I. INTRODUCTION

Remote sensing images provide essential data for observing and tracking global environmental changes [1], [2]. Change Detection (CD) can recognize the land-cover differences between multitemporal images of the same geographical area [3], [4]. Hyperspectral images (HSIs) provide high spectral resolution and abundant surface spectral details, enabling the identification of minute land-cover variations that are often missed by other imaging modalities [5], [6]. Therefore, HSI-CD has been widely adopted in various applications, such as forestry and agricultural monitoring [7], [8], environmental interpretation [9], and disaster assessment [10]. The launch of dedicated satellites, starting with the NASA Earth Observation-1 (EO-1) and more recently, the Chinese Gaofen-5, the Italian Space Agency PRISMA, and the German EnMAP, has greatly expanded the global availability of HSIs. Accordingly, HSI-CD has gained growing research focus in recent years.

HSI-CD is divided into two main tasks, which include binary change detection (BCD) and multi-class change detection (MCD) [5]. Most of past research mainly focused on BCD, which merely distinguishes whether the pixels have changed or not in bitemporal HSIs [11]. On the contrary, HSI-MCD not only locates changed areas but also determines their specific change types, which aligns well with critical applications [12], [13].

Earlier studies mainly concentrated on HSI BCD. There are many classical methods, including change vector analysis (CVA) [14], principal component analysis (PCA) [15], and slow feature analysis (SFA) [16]. However, these methods are threshold-dependent, sensitive to noise and environmental variations, and show limited generalization. They are generally restricted to binary detection and struggle to handle complex multi-class changes. In the field of weakly supervised learning, a novel approach pioneered weakly supervised semantic mapping in remote sensing by introducing bilateral filtering [17], significantly reducing annotation costs [18].

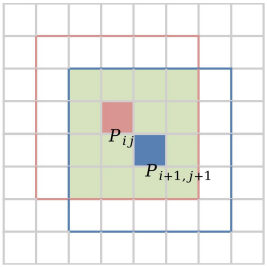


Fig. 1. Overlapping pixels cause large computation redundancy in patch-based local learning frameworks.

Deep learning methods can extract more discriminative features, achieving superior performance in HSI-MCD [19], [20]. Existing methods are roughly categorized into patch-based local learning [21], [22] and patch-free global learning frameworks [23], [24]. Patch-based methods take as input patch pairs with a fixed size, exploiting local spatial context information within each patch. In contrast, patch-free methods directly process HSIs, maximizing the exploitation of the global spatial context information across the entire HSI.

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Lifeng Wang and Bosheng Wei are with the School of Technology, Beijing Forestry University, Beijing, 100083, China (e-mail: wanglifeng_2016 @163.com).

Maofan Zhao and Yuanxin Ye are with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu, 611756, China (e-mail: mfzhao@swjtu.edu.cn, yeyuanxin@home.swjtu.edu.cn).

Lorenzo Bruzzone is with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy.

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Convolutional neural network (CNN) has long served as the fundamental architecture for HSI-CD. Early CNN-based methods are designed to model the local spectral-spatial features of bi-temporal HSIs [25]. Subsequent improvements incorporated autoencoders [26] and difference learning [27] to enhance change feature representation and improve the robustness of change discrimination in complex scenarios. To better characterize multi-scale change regions, hierarchical and multi-scale CNN were introduced, integrating spectral-spatial feature across varying receptive fields [28]. Lightweight 3D -CNN were also applied to preserving joint spectral-spatial modeling capability [29]. However, these methods follow patch-based local learning framework, suffering from inherent limitations :1) large, overlapped patches among adjacent pixels result in redundant computation, thus increasing the processing time (see Fig. 1); 2) the limited patch size restricts the receptive field and prevents the modeling of long-range dependence across the entire HSI (see Fig. 2).

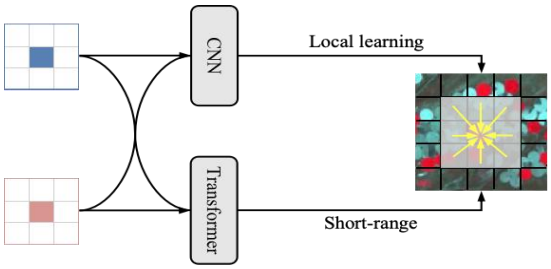


Fig. 2. Limited patch size constrains receptive fields to local neighborhoods, hindering global context modeling in patch-based frameworks.

To model global dependencies, Transformer-based methods have been introduced for HSI-CD [30]. Early adaptations used Transformer encoders to process spectral-spatial token sequences, demonstrating stronger context modeling than CNNs (see Fig. 3(a)) [31]. Later works integrated positional embedding to exploit spectral-spatial-temporal information [32]. Other papers, such as CSANet [33], employed cross-temporal interactive symmetric attention to fuse differential features. However, besides the inherent limitations of patch-based framework, the self-attention mechanism in Visual Transformers (ViT) incurs quadratic computational complexity in sequence length (see Fig. 3(b)), which becomes prohibitive for handling high-dimension HSI. This drawback leads to excessive computational and memory costs on large-scale HSI, limiting practical applications.

Recently, the selective state space model (SSM) has attracted considerable attention, offering the capability to model long-range dependency, with linear complexity [34]. Its vision counterpart, Vision Mamba [35], spurred the adoption of SSMs in visual tasks. In HSI analysis, models such as MambaHSI [36] and HyperMamba [37] explored spectral-spatial sequencing within SSM transitions. Subsequent architectures like 3DSS-Mamba [38] and IGroupSS-Mamba [39] incorporated hierarchical or multi-branch designs for multi-scale representation and long-range dependency propagation. More recent models, including S2Mamba [40], SpiralMamba [41], and MambaMoE [42], further improve modeling flexibility and robustness through advanced spatial scanning strategies. However, as shown in Fig. 3(c), although Mamba partially alleviates the local modeling limitations of CNNs and the high computational cost of Transformers, its linear scanning serializes 2D spatial features into 1D sequences, which may separate spatially adjacent pixels and weaken short-range dependency modeling. Moreover, Overlapping patches introduce redundant computation, and the fixed patch size restricts the receptive field, limiting the capture of global long-range dependencies across the entire HSI.

Patch-free global learning framework directly processes the entire HSI, thereby avoiding redundant computations and leveraging global long-range dependencies. For instance, FPGA [23] and SFFGL [24] construct patch-free global learning frameworks to reduce the redundant computation caused by overlapping patches and to fully exploit the global spatial context of the whole HSI. However, these methods mainly rely on fully convolutional operations, whose feature representation capability is still constrained by the limited receptive fields of convolution kernels, as shown in Fig. 3(a). Although Mamba has been introduced to overcome the limitation of local receptive fields and strengthen long-range dependency modeling [43], its inherent scanning strategy neglects local short-range dependencies, causing it to overlook fine-grained textures, edges, and structural details during the subsequent global scanning process.

To address the above issues, we propose a patch-free global-local fast inference framework. The main contributions of this article are summarized as follows:

1) We propose FreeViM, a novel patch-free global-local fast inference framework for HSI-MCD. FreeViM supports efficient global context sequence modeling, extracts locally spatial details, and their interactive integration.

2) We design a patch-free VMamba network that conducts linear selective scanning across the entire HSIs to model global context and enable patch-free inference. This architecture overcomes the limited receptive field of fixed-size patches, establishes long-range global dependencies and avoids redundant computation caused by overlapping patches in traditional patch-based methods, thereby improving inference speed.

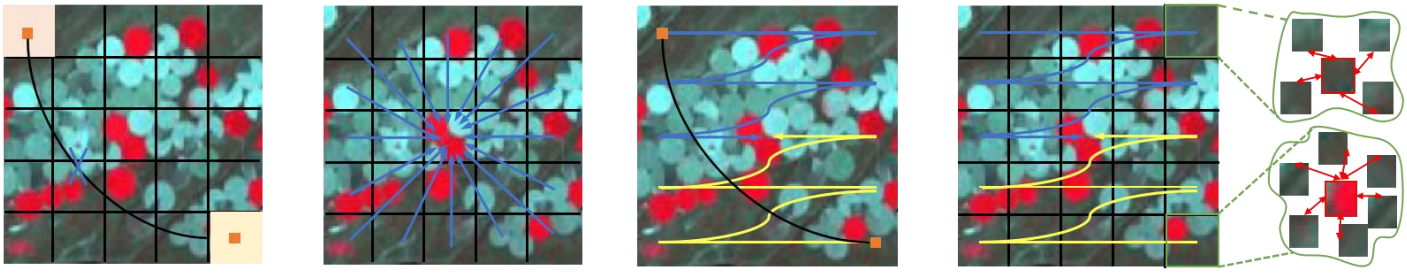
3) To address VMamba's limitation in capturing short-range dependencies, we introduce a local context clustering module that dynamically aggregates local spatial context within key regions.

4) We further propose an adaptive global-local feature interaction strategy, which employs an adaptive weighting mechanism to dynamically fuse global VMamba features and local cluster features. This allows the model to jointly capture long-range contexts and short-range details, thereby enhancing feature representation capability.

The rest of this article is organized as follows. Section II reviews related work. Section III describes the proposed method in detail. Section IV presents experimental results and parameter analyses. Finally, Section V concludes this article.

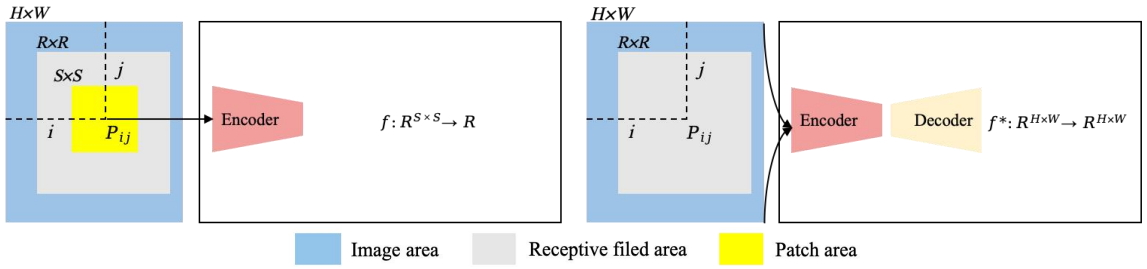
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(a) CNN (b) ViT (c) VMamba (d) Ours

Fig. 3. Illustration of feature interaction mechanisms in HSI-CD. (a) CNN treat images as grids and extract local features via sliding convolutions. (b) ViT models global relations by treating images as token sequences. (c) VMamba achieves a global receptive field with linear complexity through cross-scan integration. (d) Our method combines cross-scan with context clustering, treating images as point sets and dynamically grouping local points to capture contextual information.



(a) Patch-based local learning framework (b) Patch-free global learning framework

Fig. 4. Patch-based local learning framework and Patch-free global learning framework.

II.RELATED WORK

A. Patch-Free Global Learning Framework

As mentioned, deep learning-based methods in HSI-CD primarily follow patch-based local learning framework, such as SSA-SiamNet [21], S2Mamba [40], HybridSN [44]. Among these methods, prevalent CNN-based methods extract spectral-spatial features from a fixed-size HSI patch pair by learning a mapping $f: R^{S\times S} \rightarrow R$, as shown in Fig. 4(a). These input patches are generated from the original image by a fixed window of size S×S, with each patch containing (S-1)×(S-1) neighboring pixels around one central pixel. However, despite achieving certain accuracy, these methods still face two key limitations: 1) As shown in Fig. 1, the patch processing mechanism causes extensive overlap between adjacent patches, leading to redundant computations and longer processing time; 2) The inherently limited receptive field of local patches restricts ability to model long-range dependencies across the entire image, as shown in Fig. 2.

To address these issues, the patch-free global learning framework was introduced in FPGA [23] and SFFGL [24]. As shown in Fig. 4(b), these methods take the entire HSI as input through fully convolutional networks (FCNs), enabling end-to-end pixel-level prediction. This avoids redundant computation, significantly improves inference efficiency, and captures richer spatial context across the image.

The core idea of patch-free global learning framework is to replace explicit patches with the model's inherent receptive field. This eliminates redundant computation in overlapping areas while capturing broader potential spatial context [45]. Specifically, let X_1, $X_2 \in R^{H\times W\times C}$ be the bitemporal HSIs. The hierarchical sequence of the concatenated HSIs $X_{\text{con}} = [X_1, X_2] \in R^{H\times W\times 2C}$ is the input to the global learning framework. The predicted probability cube $\hat{Y}_i \in R^{H\times W\times Class}$ is formulated as:

$$\hat{Y}_i = f^*(X_{\text{con}}), \tag{1}$$

where the mapping $f^*: R^{H\times W\times 2C} \rightarrow R^{H\times W\times class}$ represents the full image segmentation network, H and W represent the spatial dimensions, and C is the number of spectral bands.

The classification loss $\mathcal{L}_{\text{CE}}$ (e.g., cross-entropy loss) is optimized via stochastic gradient descent minimize over the set of sampled positions $\mathcal{D}$. For the i-th iteration, the k-th model weight is updated as follows:

$$W_{i+1}^{(k)} = W_i^{(k)} - \eta \frac{1}{n} \sum_{p\in\mathcal{D}} \frac{\partial l(\tilde{Y}_i(p), \hat{Y}_i(p))}{\partial W_i^{(k)}}, \tag{2}$$

where p is the 2D spatial position in $\mathcal{D}$, $n = |\mathcal{D}|$, and η represents the learning rate and $\tilde{Y}_i$ is the sampled ground-truth map.

To ensure the convergence of the end-to-end training model, a global stratified sampling strategy is adopted. The core idea of this strategy is to transform all the training samples into a stochastic sequence of hierarchical samples. We randomly select the training pixels from the entire concatenated HSI X_{con} and the remaining pixels are used as the test data. The labeled training pixels are divided into different clusters with a certain ratio α. In detail, each cluster contains all categories, and the minibatch per class is determined by the parameter α. The number of corresponding clusters is 1/α and all of them form a stochastic hierarchical sequence. The hierarchical sequence of training samples is given as input to the network during the model training phase. Moreover, the sequence is randomly shuffled to obtain stochastic gradients and prevent overfitting.

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The main difference from patch-based methods lies in the training process: all the pixels participate in the forward computation, but only the sampled positions receive supervision in each iteration. This design maintains consistency between the training and testing phases, as both rely on a single forward pass. The one-shot inference thus accelerates change detection. Moreover, the patch-free global learning framework enables more effective use of spatial context, thereby enhancing its potential to improve the change detection performance.

Recently, the patch-free global learning framework has been introduced for HSI-CD [24]. However, despite this progress, most existing approaches remain CNN-based. Due to the limited receptive field of CNNs, deep stacking is required to capture long-range dependencies, which greatly increases computational cost. Therefore, a key yet unresolved challenge is how to further enhance a model's ability to efficiently capture global and long-range context while retaining the advantages of patch-free processing.

B. Mamba and Visual Mamba

The Mamba architecture [34], derived from state-space models, has been investigated as a computationally efficient alternative to Transformer-based sequence modeling. Traditional SSMs describe continuous-time dynamical systems and can be discretized for sequence processing. Building on this formulation, Mamba makes several SSM parameters input-dependent, enabling selective propagation of sequence information while retaining linear computational complexity [46]:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \tag{3}$$

$$y(t) = \mathbf{C}h'(t) + \mathbf{D}x(t), \tag{4}$$

where $\mathbf{A} \in R^{d\times d}$, $\mathbf{B}$, $h'(t)$, $h(t) \in R^{d\times 1}$, $\mathbf{C} \in R^{1\times d}$, and $\mathbf{D} \in R^{1\times d}$. To integrate SSM into modern deep learning pipelines, zero-order hold discretization [47] converts it into the discrete S4 formulation:

$$h_i = \bar{\mathbf{A}}h_{i-1} + \bar{\mathbf{B}}x_i, \tag{5}$$

$$y_i = \mathbf{C}h_i + \mathbf{D}x_i, \tag{6}$$

where i denotes the position of the current state within the sequence, $\bar{\mathbf{A}} = exp(\Delta\bar{\mathbf{A}})$, $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(exp(\Delta\mathbf{A} - I))\cdot\Delta\mathbf{B}$. From Eqs. (5) and (6), we can infer that sequence information can achieve global interaction with linear complexity through the propagation of h_i.

Building upon its foundation, Mamba elevates the functionality of $\mathbf{B}$, $\mathbf{C}$, and Δ by redefining them as functions of x. This transformation empowers Mamba to efficiently select data in an input-dependent manner. The selective SSM (S6) endows Mamba with the capability to dynamically adjust its mappings in response to fluctuating inputs, ensuring that the model achieves promising results across multiple visual tasks [35], [48]. In [36], Mamba demonstrated its advantages in remote sensing semantic segmentation.

Recently, several HSI-CD methods based on the Mamba architecture have been proposed, and their superior performance has been empirically validated [40]. In GDAMamba [43], a global difference-aware Mamba framework is applied to the entire image, integrating a global context difference-aware module, temporal-spectral difference enhancement, and an adaptive gated fusion mechanism, enabling the modeling of pixel-wise long-range dependencies on the difference map. However, although these methods introduce Mamba within a patch-free framework, they tend to disrupt local spatial structures while performing global modeling.

C. Context Clustering

Context Clustering [49] treats an image as a set of feature points and dynamically groups points according to their feature similarity and spatial context. Unlike convolutional operations on regular grids or token-wise self-attention, context clustering aggregates information through data-dependent point-to-center assignment and has been investigated in several visual representation tasks. Its fundamental feature aggregation formulas are:

$$S_{i,j} = \alpha \cos\theta_{i,j} + \beta, \tag{7}$$

$$c_j' = \frac{1}{M}\sum_{i=1}^{N}\sigma(S_{i,j})p_i, \tag{8}$$

where $p_i \in R^d$ denotes the i-th feature point in the input point set $p_{\text{input}} \in R^{N\times d}$, and $c_j \in R^d$ represents the j-th learnable cluster center ($j \in \{1, 2..., k\}$). M is the sum of attention weights of all samples with respect to the j-th cluster, serving as a normalization factor for the weighted average. N is the total number of pixels. α and β are learnable scaling and shifting parameters, and $S_{i,j}$ represents the cosine similarity between the feature point and the cluster center. To be effectively utilized in deep learning applications, context clustering aggregates contextual features into the center point c_j' via Eq. (8), where $\sigma(\cdot)$ is the Sigmoid activation function used to control the weight of feature assignment, and M is the normalization factor [50].

As can be inferred from Eqs. (7) and (8), through the propagation of c_j', the discrete point set information can achieve global-range feature interaction and context modeling with lower computational complexity.

Building upon this, contextual clustering enhances the representational capacity of the model through a feature dispatching mechanism, which reassigns the aggregated center features back to each original point:

$$p_i' = p_i + \sum_{j=1}^{k}\sigma(S_{i,j})c_j', \tag{9}$$

where k is the total number of learnable cluster center.

This transformation endows contextual clustering with the ability to efficiently select and fuse data in an input-dependent manner. The dynamic context clustering mechanism enables the model to dynamically adjust its receptive fields and feature mappings according to fluctuating inputs, ensuring that the model achieves remarkable results across multiple visual tasks [49].

Recently, several frameworks based on contextual clustering have been introduced into HSI feature extraction, achieving significant performance [51]. In [52], a spectral-spatial dynamic clustering network was proposed, which employs a dual-branch framework to simultaneously utilize local clustering and global

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spectral similarity for information aggregation. In [53], a superpixel-based context feature dispatching mechanism was designed to transform image patches into unordered point set sequences, thereby enhancing the model's capacity to represent complex local spatial topologies.

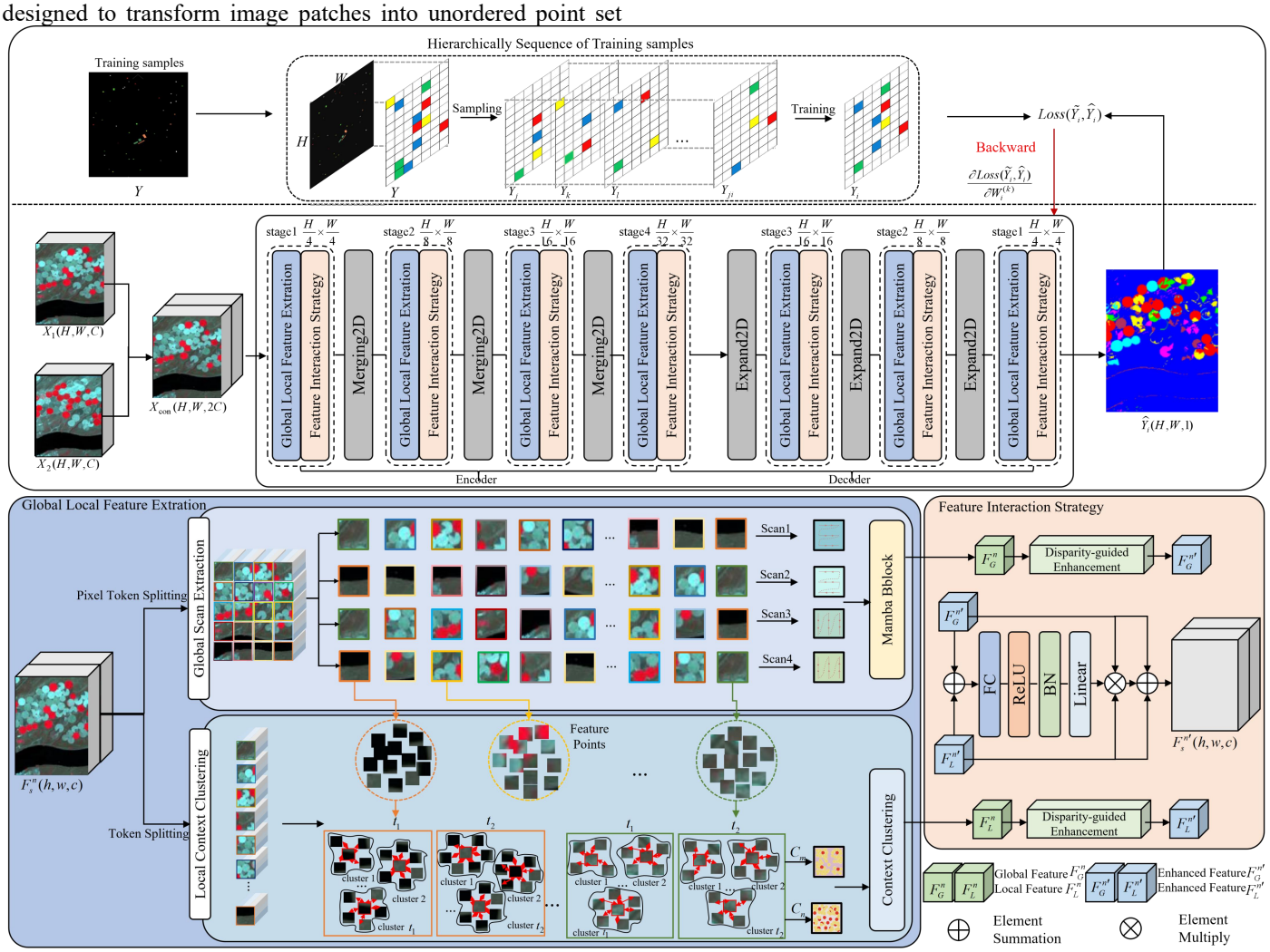


Fig. 5. Overview of the proposed FreeViM framework for HSI-MCD.

III. METHODOLOGY

As mentioned in previous section, the main bottlenecks in current HSI-MCD methods that restrict accuracy and efficiency are: 1) patch-based methods lead to redundant computation on highly overlapping areas between patches; 2) fixed patch sizes constrain the receptive field, limiting long-range global dependencies across the entire HSIs; 3) existing VMamba-based methods lose short-range dependencies by flattening spatial tokens and rely on fixed scans, which limits dynamic context and long-range interactions. To address these issues, we propose a FreeViM framework.

As shown in Fig. 4, FreeViM consists of three key components: 1) a patch-free VMamba network, enables global context modeling while avoiding redundant computation caused by overlapping patches; 2) a local context clustering module dynamically captures fine-grained local spatial details; 3) an adaptive global-local feature interaction strategy fuses long-range context with short-range details. This section presents an overview of FreeViM, followed by detailed descriptions of FreeViM, the local context clustering module, and the adaptive global-local feature interaction strategy.

A. Patch-Free Vision Mamba (FreeViM)

FreeViM learns a mapping $f^*: R^{H\times W} \to R^{H\times W}$ for HSI-MCD, as illustrated in Fig. 5. The input is the concatenation of bitemporal HSIs, denoted as $X_{\text{con}} = [X_1, X_2] \in R^{H\times W\times 2C}$. Channel-wise concatenation preserves the temporal-specific spectral signatures of both HSIs without premature differencing, while their spatial alignment allows subsequent layers to learn cross-temporal spectral-spatial correlations for distinguishing genuine changes from simple spectral variations.

First, shallow features $F_s^1 \in R^{H/4\times W/4\times c}$ are extracted using 2D convolution, batch normalization (BN), and Gaussian Error Linear Unit (GELU) activation. Then, these shallow features are passed through a four-stage encoder, where each stage comprises a Global Local Feature Extraction and Feature Interaction component.

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Between stages, Merging2D down-sampling reduces spatial resolution while increasing channel depth, resulting in feature maps F_s^n of size $h\times w$, where $h\times w$ corresponds to $H/4\times W/4$, $H/8\times W/8$, $H/16\times W/16$, and $H/32\times W/32$, with corresponding channel dimensions of c=[C, 2C, 4C, 8C]. The decoder follows a symmetric four-stage structure, using expand2D up-sampling to restore spatial resolution, with channel dimensions [8C, 4C, 2C, C].

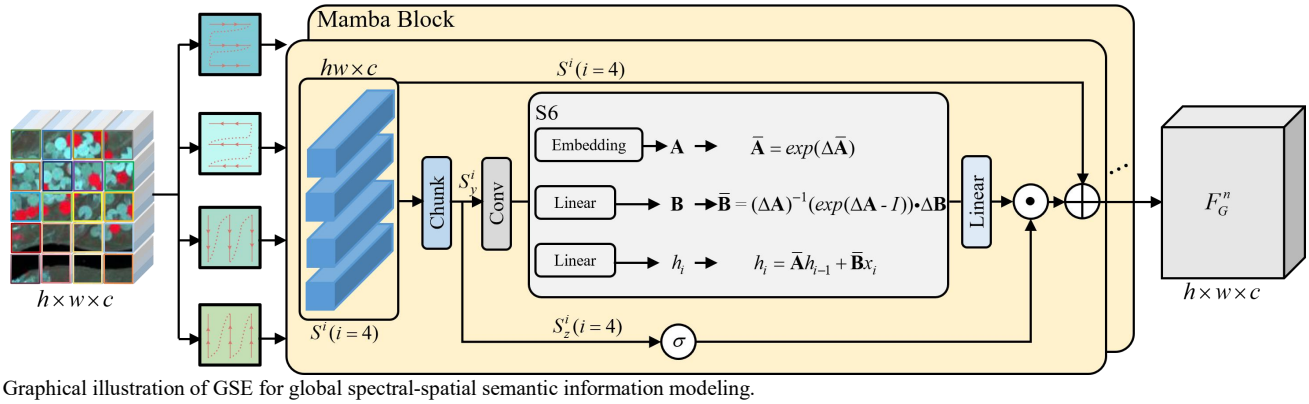


Fig. 6. Graphical illustration of GSE for global spectral-spatial semantic information modeling.

Within each Global Local Feature Extraction module, the input is first partitioned into non-overlapping patches via token splitting. Then, these are processed in parallel by global scan extraction (GSE) and LCC extraction to capture global-local features. The GSE branch utilizes a VMamba block to model global-local spatial correlations, yielding global feature F_G^n in n-th stage, while the LCC branch produces local features F_L^n in n-th stage:

$$F_G^n = \mathrm{GSE}(\mathrm{GN}(\mathrm{SSM}(F_S^n))), \tag{10}$$

$$F_L^n = \mathrm{LCC}(\mathrm{GN}(\mathrm{SSM}(F_S^n))), \tag{11}$$

where $F_G, F_L \in R^{h\times w\times c}$ and $h\times w\times c$ denotes sizes of feature maps at different stages. GN(·) represents Group Normalization, and SSM(·) denotes the State Space Model. The feature processing operations of GSE(·) and LCC(·) are detailed in Sections III-B and III-C, respectively.

Then, F_G^n and F_L^n are enhanced through a feature enhancement block, resulting in $F_G^{n'}, F_L^{n'} \in R^{h\times w\times c}$. These are then adaptively fused via global-local feature interaction fusion strategy to integrate the global-local feature detailed in Section III-D.

Finally, an output module comprising global average pooling, channel-wise convolution, and argmax produces pixel-wise class probabilities $Prob \in R^{H\times W\times K}$, K is the number of change classes.

B. Global Scan Extraction (GSE)

The FreeViM network enabling unified modeling of long-range global dependencies and short-range local details through integrated global-local spectral-spatial information. GSE and LCC operate in parallel, performing global Mamba linear scanning and local cluster feature sharing, respectively.

Mamba processes input in a causal manner, which is well-suited to 1D language sequence but inherently mismatched with the 2D structure of images [35]. Simply flattening the image fails to capture spatial context. To address this, VMamba introduces a 2D cross-scanning mechanism [48]. As shown in Fig. 6, the HSI is traversed pixel-wise, and selective scanning is performed along four directions (↘, ↖, ↙, ↗), serializing the spatial data into 1D sequences $\{S^1, S^2, S^3, S^4\} \in R^{hw\times c}$. This design enables each pixel to integrate multi-directional global context while maintaining linear computational complexity. These sequences are processed by the Mamba as follows:

$$S_y^i, S_z^i = \Phi_{chunk}(\mathrm{linear}(\mathrm{LN}(S^i))), \tag{12}$$

where $\Phi_{chunk}(\cdot)$ denotes the splitting of the input feature along the channel dimension into multiple branches with different semantic meanings. The content branch S_y^i propagates projected spectral-spatial features through selective state-space recurrence to capture long-range contextual dependencies, while the gating branch S_z^i provides input-dependent modulation to adaptively control their contribution to the output, respectively. We can write:

$$h_t = A_t \odot h_{t-1} + B_t, \tag{13}$$

$$\bar{S}_y^i = C_t h_t + D_t \odot \sigma(S_y^i), \tag{14}$$

where A_t, B_t, C_t, and D_t are discrete SSM parameters, and h_t is the hidden state at step t [34]. The output feature S_{out} can be obtained as:

$$S_{out} = \mathrm{Reshape}(\mathrm{linear}(\mathrm{LN}(\bar{S}_y^i)) \odot \sigma(S_z^i) + S^i), \tag{15}$$

where $\bar{S}_y^i$ is the intermediate state. By decoupling the input into content and gating streams, VMamba performs input-dependent selective state updates that propagate informative tokens while suppressing irrelevant or noisy information. This design enhances both model flexibility and stability, all with linear computational complexity.

Finally, the globally scanned features are passed through an integration unit consisting of a linear layer, a normalization, and the GELU activation function. The final global feature representation F_G^n is obtained as:

$$F_G^n = \mathrm{GELU}(\mathrm{LN}(\mathrm{linear}(S_{out}))). \tag{16}$$

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C. Local Context Clustering (LCC)

Instead of relying on convolution or attention mechanisms for local feature extraction, we adopt the context clustering algorithm [49] to capture local spatial context information more adaptively. In the context clustering layer, the feature map is treated as an unordered set of data points and dynamically grouped into clusters within local windows. To avoid the high computational cost of global clustering, we partition the feature map into non-overlapping local windows and perform clustering independently within each window [46].

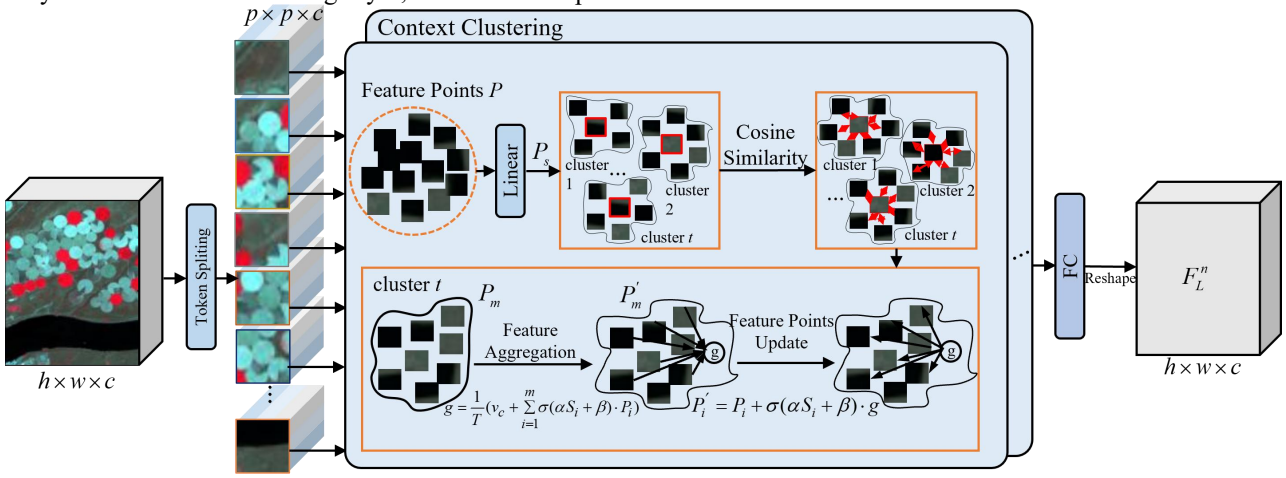


Fig. 7. Graphical illustration of LCC for local spectral-spatial semantic information modeling.

As illustrated in Fig. 7, after the token splitting layer, the input image F_s^n is divided into non-overlapping windows of size $p \times p \times c$. These windows are then flattened into a set of data points $P \in R^{p^2 \times d}$, where d denotes the feature dimension.

To compute the similarity between points, which guides the initialization of cluster centers and the assignment of points, the original feature points P are first projected into the similarity space, resulting in $P_s \in R^{p^2 \times d'}$, where d' is the dimension of the projected space. Within each local window, we initialize t cluster centers by averaging the features of their k-nearest neighbors [54], which preserving homogeneous regional consistency while remaining sensitive to boundaries and small change regions, Compared with random or learnable centers. Pairwise cosine similarity between these t centers and all points P_s is then computed, yielding a similarity matrix $S \in R^{p^2 \times t}$. Each point is assigned to its most similar center, forming t clusters. Note that clusters may contain varying numbers of points and, in rare cases, may also be empty.

Let us consider a semantic cluster $P_m \in R^{m \times d'}$ containing m points with corresponding similarities $S_i \in R^m$ (a subset of S). Before feature aggregation, these points are projected into a value space through a learnable linear mapping $\phi_v[\cdot]$:

$$P_m^v = \phi_v(P_m) = [v_1, ..., v_m] \in R^{m \times d_v}, \tag{17}$$

where d_v denotes the value-space dimension and v_i is the value representation of the i-th point. A value center [55] $v_c \in R^{d_v}$ is further computed in this space. The aggregated cluster feature $g \in R^{d'}$ is then formulated as:

$$g = \frac{1}{T}(v_c + \sum_{i=1}^{m} \sigma(\alpha S_i + \beta) \cdot v_i), \tag{18}$$

where $T = 1 + \sum_{i=1}^{m} \sigma(\alpha S_i + \beta)$, α and β are learnable scalars for scaling and shifting the similarities. The inclusion of the value center v_c enhances numerical stability and emphasizes locality.

Subsequently, the aggregated cluster feature g is adaptively redistributed to each point using point-specific similarity weights and combined with the original point features, thereby preserving individual details while avoiding excessive intra-cluster smoothing. Each point P_m is updated as:

$$P_i' = P_m + \sigma(\alpha S_i + \beta) \cdot g, \tag{19}$$

where updates the cluster center by applying nonlinear mapping and dynamic modulation to the clustering similarity score S_i, thereby generating adaptive weights to enhance the response to highly similar samples and improve feature aggregation capability.

It should be noted that LCC is intentionally designed as a local window modeling rather than an explicit cross-window propagation mechanism. The non-overlapping partition substantially reduces the computational cost of clustering, but may introduce contextual isolation near window boundaries. In FreeViM, this limitation is compensated by the parallel GSE branch, whose multidirectional selective scanning propagates information over the entire feature map, together with the hierarchical Merging2D/Expand2D operations across successive stages. Therefore, LCC focuses on fine-grained intra-window spatial structures, whereas GSE provides complementary inter-window and long-range contextual interactions [56].

D. Global-local feature interaction strategy

The global-local feature interaction strategy dynamically integrates global features from VMamba and local features from context clustering. Global features capture long-range spectral-spatial dependencies across the entire image with linear computational complexity. Local features preserve

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fine-grained textures and structural details that global scanning may overlook.

As shown in Fig. 8, the interaction process consists of two sequential stages: disparity-guided enhancement and adaptive fusion.

*1)**Disparity-guided enhancement.***

Global features F_G^n and local features F_L^n are first combined by element-wise addition to preserve their shared semantics, while subtraction highlights complementary discrepancies; the resulting disparity information is then processed by GAP-Sigmoid to generate adaptive modulation weights that enhance complementary cues and suppress redundant responses. These weights selectively amplify complementary information in each branch, enhancing broad contextual patterns from Mamba and localized boundary variations from clustering [57]. The enhanced features are formulated as:

$$F_G^{n\prime} = F_G^n \odot (1+\sigma(\mathrm{GAP}(F_G^n - F_L^n))), \tag{20}$$

$$F_L^{n\prime} = F_L^n \odot (1+\sigma(\mathrm{GAP}(F_L^n - F_G^n))), \tag{21}$$

where $\mathrm{GAP}(\cdot)$ represents global average pooling. This mechanism draws from disparity-aware and guided fusion designs in contemporary Mamba architecture.

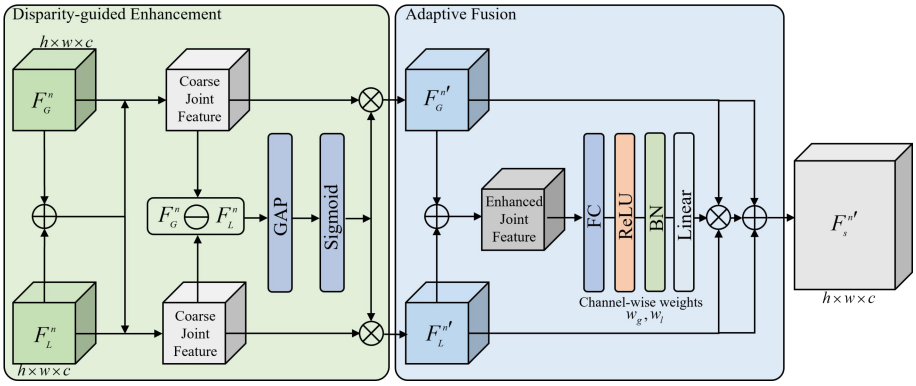


Fig. 8. Structure of the proposed global-local feature interaction strategy.

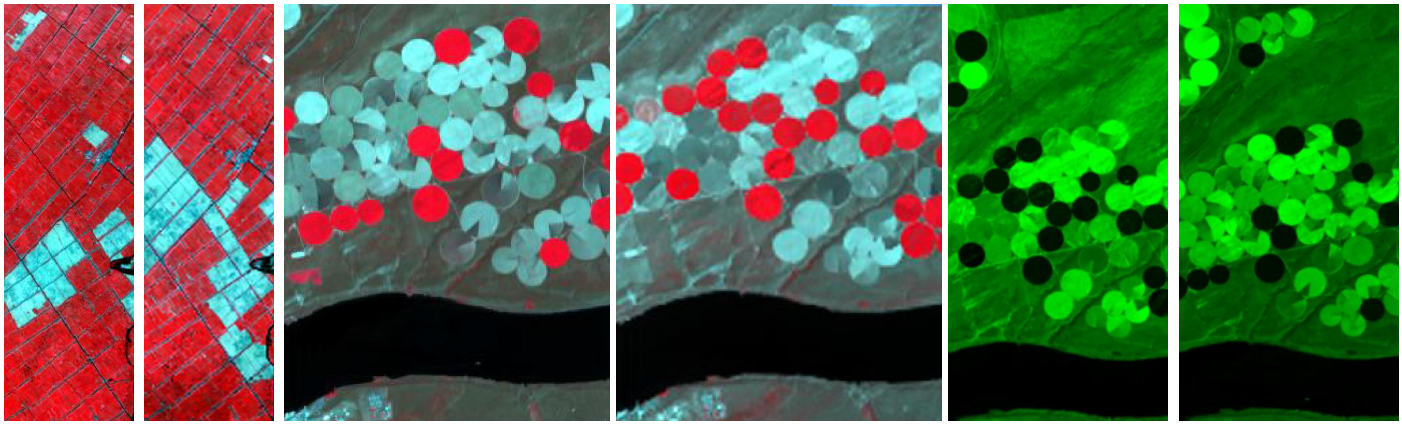
2)Adaptive fusion.

The enhanced features undergo content-aware integration. A preliminary joint representation is compressed through ReLU and batch normalization layer [58] to derive channel-wise importance weights w_g and w_l. The final fused representation is computed with a residual connection:

$$F_s^{n\prime} = w_g \odot F_G^{n\prime} + w_l \odot F_L^{n\prime} + (F_G^{n\prime} + F_L^{n\prime}), \tag{22}$$

$$w_g, w_l = \mathrm{ReLU}(\mathrm{BN}(\mathrm{GAP}(F_L^n - F_G^n))) \tag{23}$$

By progressing from explicit disparity highlighting to learnable weighted integration, the strategy produces a unified representation that is both globally coherent and locally detailed. This design overcomes the loss of fine structures in pure global modeling and the restricted receptive field in pure local modeling, while retaining the efficiency of the FreeViM network.



(a) (b) (c) (d) (e) (f)

Fig. 9. The three HSI datasets acquired by the Hyperion Sensor used in the experiments (bands 33, 22 and 11 used for the RGB comparison). (a) and (b) are Yancheng images acquired on May 3, 2006 and April 23, 2007. (c) and (d) are Umatilla images acquired on May 1, 2004 and May 8, 2007. (e) and (f) are Hermiston images acquired in 2004 and 2007.

E. Loss Function

To optimize the network, FreeViM adopts a weighted cross-entropy loss [24] that enhances robustness against class imbalance and spatial heterogeneity. The loss is computed pixel-wise, with an adaptive spatial weight map that emphasizes important samples and reduces the bias toward dominant classes. The overall loss is expressed as:

$$\mathcal{L}_{\mathrm{CE}} = \frac{1}{N_{valid}} \sum_{i \in \mathcal{V}} w_i \cdot \ell_{\mathrm{CE}}(p_{\mathrm{class}}, y_i), \tag{23}$$

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where v denotes the index set of all valid pixels, N_{valid} represents the total number of valid pixels, w_i is the adaptive weight assigned to pixel, p_{class} indicates the predicted class probability distribution at that location, and y_i corresponds to the ground truth label.

At the pixel-wise loss computation level, the cross-entropy term for each location measures the discrepancy between predictions and ground truth labels through the negative log-likelihood, expressed as:

$$\ell_{\text{CE}}(p_{\text{class}}, y_i) = \sum_{e=1}^{E} (y_i = e) \log(p_{\text{class}}, e) \ . \tag{24}$$

To mitigate class imbalance, the adaptive weight w_i is computed via either inverse class frequency $w_i = 1/\sqrt{y_i}$ (down-weighting dominant background classes to enhance minority change detection) or focal-style weighting $w_i = (1 - p_{\text{class}}, y_i)^{\gamma}$ (focusing training on hard-to-classify boundary pixels via the focusing parameter γ), thereby improving robustness against spatial heterogeneity.

IV.EXPERIMENTS

A. Datasets Description

To evaluate the effectiveness of the proposed method, experiments were conducted on three HSI benchmark datasets.

All datasets were acquired by the Earth Observing-1 (EO-1) satellite equipped with the Hyperion sensor. The captured HSIs have a spatial resolution of approximately 30 m and a spectral resolution of 10 nm. Fig.9 shows the false color images from the Yancheng, Umatilla, and Hermiston datasets.

The Yancheng dataset, which includes four change categories, covers farmland near Yancheng, Jiangsu province, China. The pre-change image (Fig. 9(a)) was acquired on May 3, 2006, and the post-change image (Fig. 9(b)) on April 23, 2007. The image size is 420×140 pixels with 154 spectral bands after removing noisy channels.

The Umatilla dataset covers irrigated farmland in Umatilla County, Oregon, USA. The pre-change image (Fig. 9(c)) was acquired on May 1, 2004, and the post-change image (Fig. 9(d)) on May 8, 2007. The image size is 307×241 pixels and include 154 spectral bands after removing noisy channels. Land cover types include irrigated farmland, soil, grassland, rivers, cropland, and buildings.

The Hermiston dataset covers farmland in Hermiston, Oregon, USA. The pre-change image (Fig. 9(e)) was acquired in 2004, and the post-change image (Fig. 9(f)) in 2007. The images have a size of 390×200 pixels and contain 242 spectral bands.

B. Parameter Settings and Comparison Methods

The proposed method was implemented using the PyTorch framework, and the model was trained on a single NVIDIA RTX 2070 Ti GPU. The Adam optimizer with L2 regularization (weight decay) was employed, with both the learning rate and weight decay set to 0.001. The batch size was configured as 32, and the local window was set to 5. The overall accuracy (OA), average accuracy (AA), and the Kappa coefficient are used for accuracy assessment.

To evaluate the performance of the proposed global-local framework, we compared it with seven benchmark methods, including patch-based methods (CSANet [33], HybridSN [44], MFCEN [59]), patch-free methods (FPGA [23], SFFGL [24]), a Mamba-based global learning method (GDAMamba [39]), and a Transformer-based global-local joint learning method (GLAFormer [60]). For a fair comparison, 20% of labeled pixels per change class were randomly selected for training, with the remainder used for testing.

C. Quantitative Results

Tables 1-3 summarize the average CD performance from ten independent runs. From the analysis of the tables, several conclusions can be drawn.

First, the proposed FreeViM consistently achieves the best performance among all compared methods in terms of OA, AA, and Kappa coefficient. This superiority is attributed to the global-local feature interaction within a patch-free global learning framework, which captures discriminative local information while retaining the advantages of global modeling. These results demonstrate that, for HSI MCD, the patch-free global learning paradigm is generally more effective than traditional patch-based local learning frameworks.

Second, in most cases patch-free methods outperform patch-based methods on the three datasets, particularly in terms of OA and AA. This is because patch-free methods can model long-range dependencies across the entire image without the constraints of patch size, facilitating a more global representation of change features and leading to improved change detection performance.

Third, compared with Mamba-based global learning method, integrating global-local joint learning method (GLAFormer) generally achieves better change detection performance. In the Yancheng dataset, classes with few samples (e.g., class “3”) are poorly detected by most methods. However, the proposed FreeViM reduces both missed and false detection through fusing long-range context with short-range details. In the Umatilla dataset, where complex subtle land cover changes often lead to missed detection along river boundaries, FreeViM effectively captures fine-grained global-local spectral-spatial features, yielding more discriminative representation.

In summary, the effective integration of global context and local details is crucial for improving HSI-MCD performance, and the proposed FreeViM exhibits a clear advantage in this regard.

D. Qualitative Results

Figs. 10-12 show the MCD maps produced by eight different methods and the corresponding ground-truth for the Yancheng, Umatilla, and Hermiston datasets. The white rectangles highlight regions of major discrepancies among the methods.

On the Yancheng dataset, due to the extremely limited subtle change areas for the yellow class, methods such as CSANet, GLAFormer, and MFCEN exhibit unsatisfactory detection performance for classes with few samples. In contrast, FreeViM leverages its local context clustering and global-local feature interaction strategy to precisely capture information from regions with few samples, achieving superior results. On the Umatilla dataset that includes many subtle land cover change regions and closely adjacent classes, CSANet, HybridSN, MFCEN, and SFFGL suffer significant missed detection,

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particularly near river boundaries. Benefiting from the synergistic interaction between its powerful global-local feature extraction capabilities, FreeViM detects these subtle changes more effectively than the other methods.

Overall, the CD maps generated by the proposed FreeViM appear more coherent and visually consistent across all three datasets, demonstrating superior visual quality. These observations further confirm the critical importance of global-local feature interaction for HSI-MCD task.

Table 1 Accuracy obtained by different CD methods on the Yancheng dataset (Bold values indicate the best result on each term)

Class	Patch-based				Patch-free			
	CSANet	HybridSN	MFCEN	GLAFormer	GDAMamba	FPGA	SFFGL	Ours
Class 0	97.22±0.52	98.59±0.05	98.41±0.07	97.85±0.05	98.72±0.02	97.63±0.01	97.91±0.02	**98.85±0.02**
Class 1	94.68±0.29	95.16±0.29	94.12±0.52	95.28±0.30	96.45±0.11	95.56±0.03	96.67±0.27	**96.92±0.17**
Class 2	95.53±0.14	96.84±0.08	95.15±0.19	97.41±0.22	**97.90±0.15**	94.27±0.06	97.88±0.13	97.33±0.03
Class 3	60.30±5.22	74.57±4.62	73.44±1.64	83.29±2.11	86.89±1.19	74.38±3.93	71.02±4.11	**88.52±0.92**
OA (%)	96.73±0.24	97.79±0.09	97.03±0.13	97.82±0.24	97.97±0.08	97.56±0.10	97.37±0.04	**98.22±0.06**
AA (%)	87.02±0.04	91.29±1.20	90.28±0.47	93.97±0.38	94.24±0.55	91.65±0.73	91.05±1.28	**95.33±0.60**
Kappa×100	93.17±0.17	95.37±0.19	94.15±0.17	95.76±0.11	95.73±0.15	94.78±0.05	95.05±0.11	**95.78±0.05**

Table 2 Accuracy obtained by different CD methods on the Umatilla dataset (Bold values indicate the best result on each term)

Class	Patch-based				Patch-free			
	CSANet	HybridSN	MFCEN	GLAFormer	GDAMamba	FPGA	SFFGL	Ours
Class 0	97.74±0.05	98.04±0.06	97.78±0.02	97.59±0.04	98.37±0.07	97.90±0.21	98.20±0.08	**98.70±0.06**
Class 1	86.67±0.42	86.43±0.84	82.05±0.82	81.05±0.90	**88.65±0.41**	84.57±1.53	88.45±0.87	87.19±0.32
Class 2	83.46±0.29	87.89±0.68	85.67±0.02	74.97±1.21	88.07±0.22	87.02±0.91	87.73±0.97	**88.74±0.19**
Class 3	94.41±0.48	94.83±0.22	94.02±0.03	93.83±0.30	95.83±0.05	94.58±0.40	**96.50±0.31**	95.39±0.17
Class 4	86.07±0.54	88.28±0.52	84.06±0.38	83.61±0.97	85.86±0.37	85.56±0.95	83.30±0.52	**89.04±0.05**
Class 5	96.04±0.27	96.20±0.33	95.21±0.37	95.87±0.47	95.43±0.09	94.97±0.91	91.12±0.37	**96.97±0.09**
Class 6	82.90±0.98	85.34±0.56	83.37±0.92	78.18±1.77	89.67±0.06	85.22±1.20	84.44±0.45	**87.57±0.20**
OA (%)	96.08±0.06	96.53±0.05	95.77±0.14	95.07±0.05	96.57±0.11	96.27±0.07	96.28±0.08	**96.97±0.04**
AA (%)	89.62±0.21	91.00±0.07	88.22±0.01	86.21±0.18	91.90±0.33	90.59±0.12	89.54±0.24	**92.03±0.14**
Kappa×100	89.83±0.16	91.03±0.11	89.03±0.21	87.21±0.10	91.62±0.12	90.51±0.12	90.88±0.20	**91.31±0.12**

Table 3 Accuracy obtained by different CD methods on the Hermiston dataset (Bold values indicate the best result on each term)

Class	Patch-based				Patch-free			
	CSANet	HybridSN	MFCEN	GLAFormer	GDAMamba	FPGA	SFFGL	Ours
Class 0	99.59±0.05	99.66±0.04	99.60±0.05	99.82±0.06	99.73±0.03	99.69±0.04	99.76±0.14	**99.78±0.15**
Class 1	97.76±0.10	97.97±0.07	97.82±0.03	97.45±0.12	98.22±0.06	97.57±0.08	97.79±0.07	**98.30±0.75**
Class 2	97.60±0.20	97.39±0.06	96.52±0.07	97.21±0.36	97.44±0.08	97.18±0.31	**97.91±0.09**	97.42±0.06
Class 3	90.02±0.91	84.78±1.77	94.46±0.73	89.28±1.20	90.28±1.54	93.65±0.07	81.22±3.17	**94.55±0.10**
Class 4	94.81±0.77	96.44±0.22	93.69±0.35	95.88±0.36	**97.05±0.17**	96.71±0.85	96.23±0.15	96.42±0.14
Class 5	96.30±0.70	96.29±0.10	95.79±0.22	94.99±0.05	95.82±0.12	96.23±0.60	95.28±0.10	**96.32±0.13**
OA (%)	99.23±0.06	99.39±0.02	99.31±0.04	99.24±0.18	99.39±0.04	99.37±0.06	99.32±0.07	**99.52±0.05**
AA (%)	95.94±0.34	95.66±0.37	96.42±0.06	95.89±0.01	96.41±0.33	96.84±0.07	94.61±0.65	**97.31±0.56**
Kappa×100	97.10±0.21	97.39±0.08	97.11±0.09	97.11±0.18	97.43±0.05	97.30±0.50	97.16±0.01	**97.93±0.09**

E. Computation Time

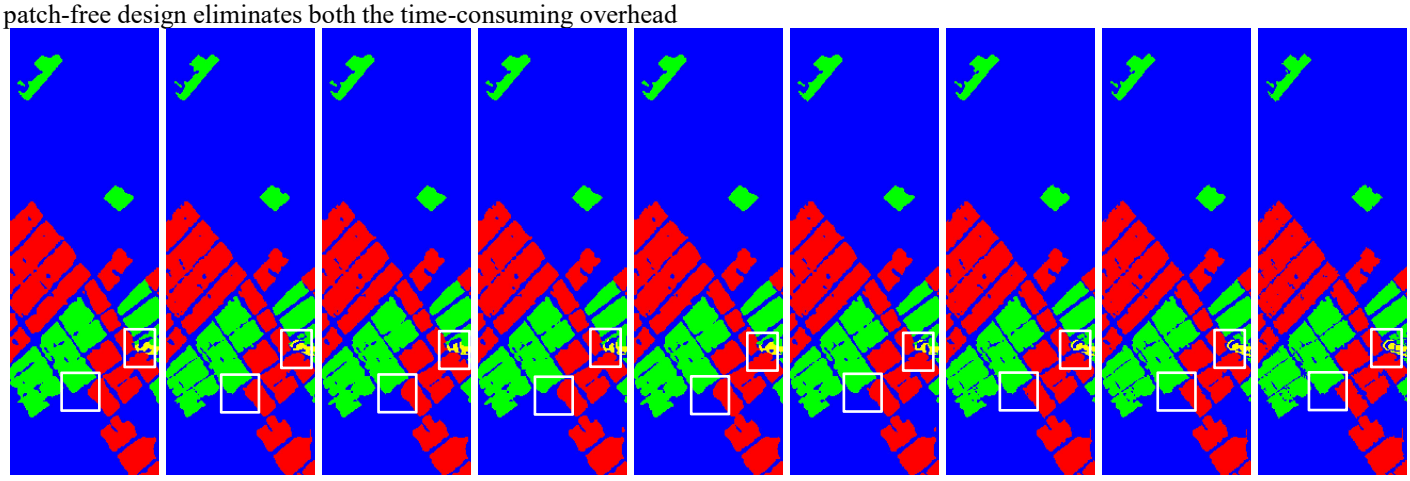
Fig. 13 compares the computation efficiency of the eight deep learning methods considered. As observed, the proposed

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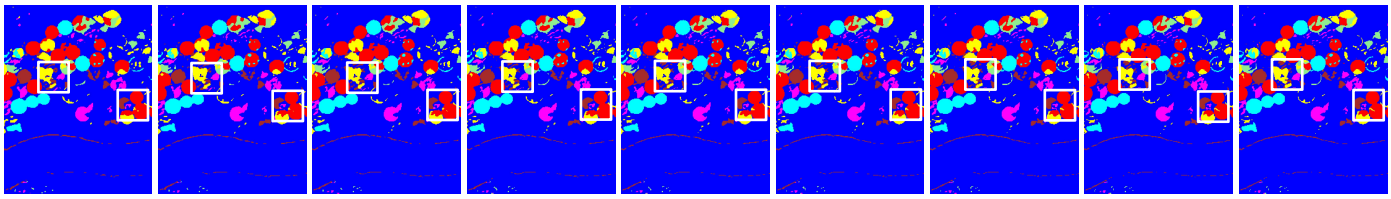
patch-free framework (i.e., FPGA, SFFGL, and FreeViM) significantly outperforms traditional patch-based methods (i.e., CSANet, HybridSN, GLAFormer, GDAMamba, and MFCEN) in terms of computation time. Building upon this, the integration of the VMamba model further accelerates inference speed.

This improvement stems from two key factors. First, the patch-free design eliminates both the time-consuming overhead of patch generation and the computation redundancy caused by overlapping patches. Second, the incorporation of VMamba leverages its efficient SSM, which maintains a global receptive field while achieving linear computation complexity. Notably, this substantial enhancement in efficiency makes the proposed method highly suitable for practical application scenarios that demand high-speed inference.



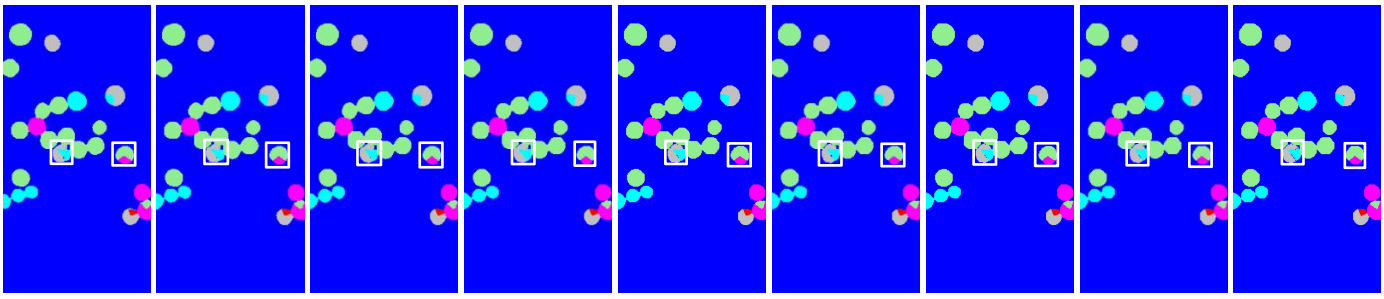
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. CD maps obtained by the different methods on the Yancheng dataset. (a) CSANet. (b) HybridSN. (c) MFCEN. (d) GLAFormer. (e) GDAMamba. (f) FPGA. (g) SFFGL. (h) Ours. (i) Ground-reference map.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 11. CD maps obtained by the different methods on the Umatilla dataset. (a) CSANet. (b) HybridSN. (c) MFCEN. (d) GLAFormer. (e) GDAMamba. (f) FPGA. (g) SFFGL. (h) Ours. (i) Ground-reference map.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 12. CD maps obtained by the different methods on the Hermiston dataset. (a) CSANet. (b) HybridSN. (c) MFCEN. (d) GLAFormer. (e) GDAMamba. (f) FPGA. (g) SFFGL. (h) Ours. (i) Ground-reference map.

F.Ablation Study

To demonstrate the effectiveness of each component in the proposed FreeViM method, we conducted ablation experiments on the three datasets. The experimental setting followed Section III-A. The baseline model follows an encoder-decoder FreeViM network architecture. Each core component was then progressively integrated, and the corresponding results are reported in Table 4.

The experimental results demonstrate a clear performance progression as each component is added. First, GSE yields a substantial improvement across all three datasets. This confirms SSM can effectively capture long-range dependencies in the entire HSIs, underscoring the foundational role of global context information in HSI-MCD. Next, augmenting the model with LCC module further increases AA scores by 1.64%, 0.80%, and 0.55% on the three datasets, respectively. These gains validate that local detail information provides essential complementary cues to global learning network. Finally, incorporating the global-local feature interaction strategy further improves performance by adaptively strengthening and integrating global context and local details. This demonstrates that jointly exploiting both global and local spectral-spatial features is critical for maximizing model effectiveness. Collectively, these progressive improvements across all datasets validate the contribution of each designed component .

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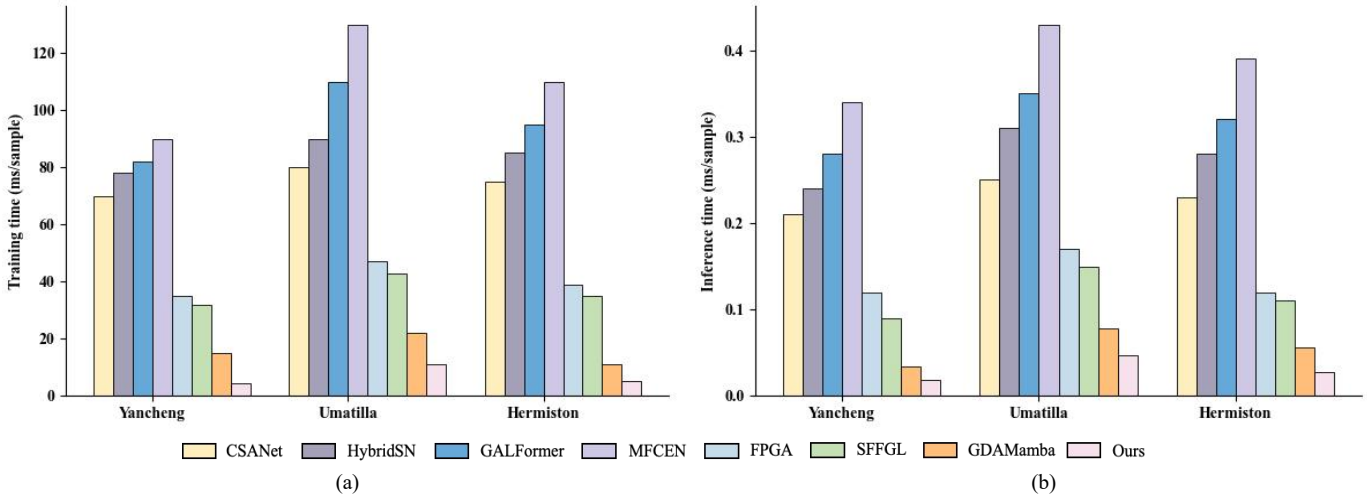


Fig. 13. Computational time of different CD methods on the three datasets. (a) Training time. (b) Inference time.

Table 4 Ablation study results with different components on the three datasets (Bold values indicate the best result in each term)

Prior			Yancheng			Umatilla			Hermiston		
GSE	LCC	Fusion	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
-	-	-	97.56±0.10	91.65±0.73	94.78±0.05	96.27±0.07	90.59±0.12	90.51±0.12	99.37±0.06	94.61±0.65	97.30±0.50
√	-	-	97.96±0.08	92.18±0.17	94.86±0.24	96.57±0.11	91.22±0.25	91.33±0.27	99.39±0.02	96.42±0.06	97.39±0.08
√	√	-	98.06±0.01	93.29±1.20	95.57±0.10	96.70±0.15	91.39±0.34	91.43±0.30	99.48±0.14	96.84±0.07	97.77±0.16
√	√	√	**98.22±0.06**	**95.33±0.60**	**95.78±0.05**	**96.97±0.04**	**92.03±0.14**	**91.31±0.12**	**99.52±0.05**	**97.31±0.56**	**97.93±0.09**

Table 5 Accuracy obtained by FreeViM with different number of cluster centers for the three datasets (Bold values indicate the best result in each term)

Cluster Center	Yancheng			Umatilla			Hermiston		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
t_1=3, t_2=15	96.27±0.07	92.18±0.17	94.68±0.05	95.57±0.10	90.51±0.12	90.51±0.12	99.37±0.06	95.66±0.37	97.16±0.01
t_1=4, t_2=20	**98.22±0.06**	**95.33±0.60**	**95.78±0.05**	**96.97±0.04**	**92.03±0.14**	**91.31±0.12**	**99.52±0.05**	**97.31±0.56**	**97.93±0.09**
t_1=5, t_2=25	97.56±0.10	93.79±1.20	95.57±0.10	96.57±0.07	91.51±0.12	91.03±0.11	99.31±0.04	96.01±0.33	97.44±0.05
t_1=6, t_2=30	96.97±0.04	93.04±0.22	94.86±0.24	96.07±0.10	90.59±0.12	90.88±0.20	99.23±0.06	94.61±0.65	97.30±0.50

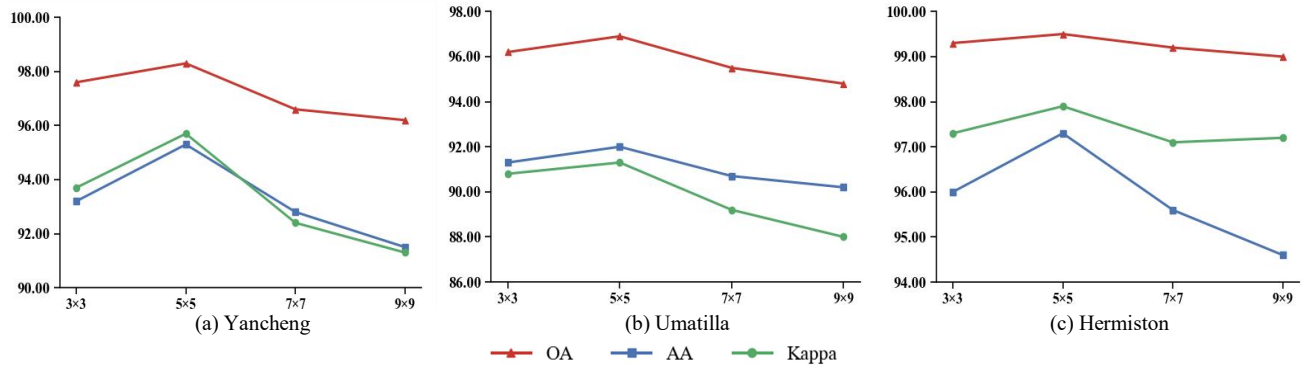


Fig. 14. Accuracy obtained by FreeViM with different local window sizes for the three considered datasets.

G. Hyperparameter Analysis

In the proposed method, several hyperparameters can affect the training model and, further, the CD results. In this section, we analyzed most crucial ones that are the local window size and the number of cluster centers.

1) Impact of Local Window Size

In the local context clustering process, the size of the local window determines the amount of spatial context captured around each center pixel. As shown in Fig. 14, we evaluated four window sizes (3×3 , 5×5 , 7×7 , and 9×9) on the three datasets. The window size of 5×5 yielded the highest OA, AA,

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and Kappa scores across all datasets, outperforming the other sizes considered. This is because a 5×5 window strikes an optimal balance between local feature aggregation and spatial detail preservation. Therefore, 5×5 is selected as the best local window size for our framework.

2) Impact of Number of Cluster Centers

The number of cluster centers in the local context clustering module is intrinsically linked to the scale and density of land cover change regions. To effectively handle multi-scale change detection, we evaluate two sets of center configurations: $t_1 \in \{3,4,5,6\}$ for large-scale contiguous changes (e.g., extensive vegetation shifts) and $t_2 \in \{15,20,25,30\}$ for subtle scattered changes (e.g., narrow road variations). In our framework, the clustering process for all three datasets simultaneously incorporates both large-scale centers t_1 and fine-grained centers t_2 to capture diverse spatial structures. A lower t_1 preserves regional coherence and prevents over-fragmentation of large areas, while a higher t_2 ensures that fine details are not overlooked. As shown in Table 5, the experimental results demonstrate that the optimal balance is achieved when t_1=4 and t_2=20, allowing the model to adaptively represent both macroscopic and microscopic land cover changes.

V. Conclusion

In this paper, we have proposed FreeViM, an efficient framework designed to address the inherent limitations of current HSI-MCD methods. Departing from traditional patch-based processing, FreeViM employs a patch-free VMamba network architecture that captures long-range global dependencies across the entire hyperspectral image while avoiding the redundant computations associated with overlapping patches. To address the deficiencies of the standard VMamba architecture in capturing short-range dependencies, we developed a dynamic local context clustering module to extract fine-grained spatial detail features. Furthermore, this study introduces an adaptive global-local feature interaction strategy, effectively achieving the fusion of global context and local details. Experimental results on three benchmark datasets (i.e., Yancheng, Umatilla, and Hermiston) demonstrate that FreeViM achieves state-of-the-art levels in both detection accuracy and computational efficiency, significantly reducing the rate of missed detections, especially when processing complex and subtle land-cover changes. Experiments on the Yancheng, Umatilla, and Hermiston datasets show that FreeViM achieves maximum gains of 0.69 percentage points in OA on Umatilla, and 1.10 percentage points in AA and 0.73 points in Kappa×100 on Yancheng over the strongest competing baselines. These results demonstrate that FreeViM effectively balances global-context modeling, local-detail representation, and efficient patch-free inference for HSI-MCD. Despite these advantages, FreeViM still has an architectural limitation: the local context clustering branch operates within fixed non-overlapping windows, without explicitly modeling local interactions across window boundaries, which may constrain the continuity of fine-grained spatial structures near window boundaries.

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